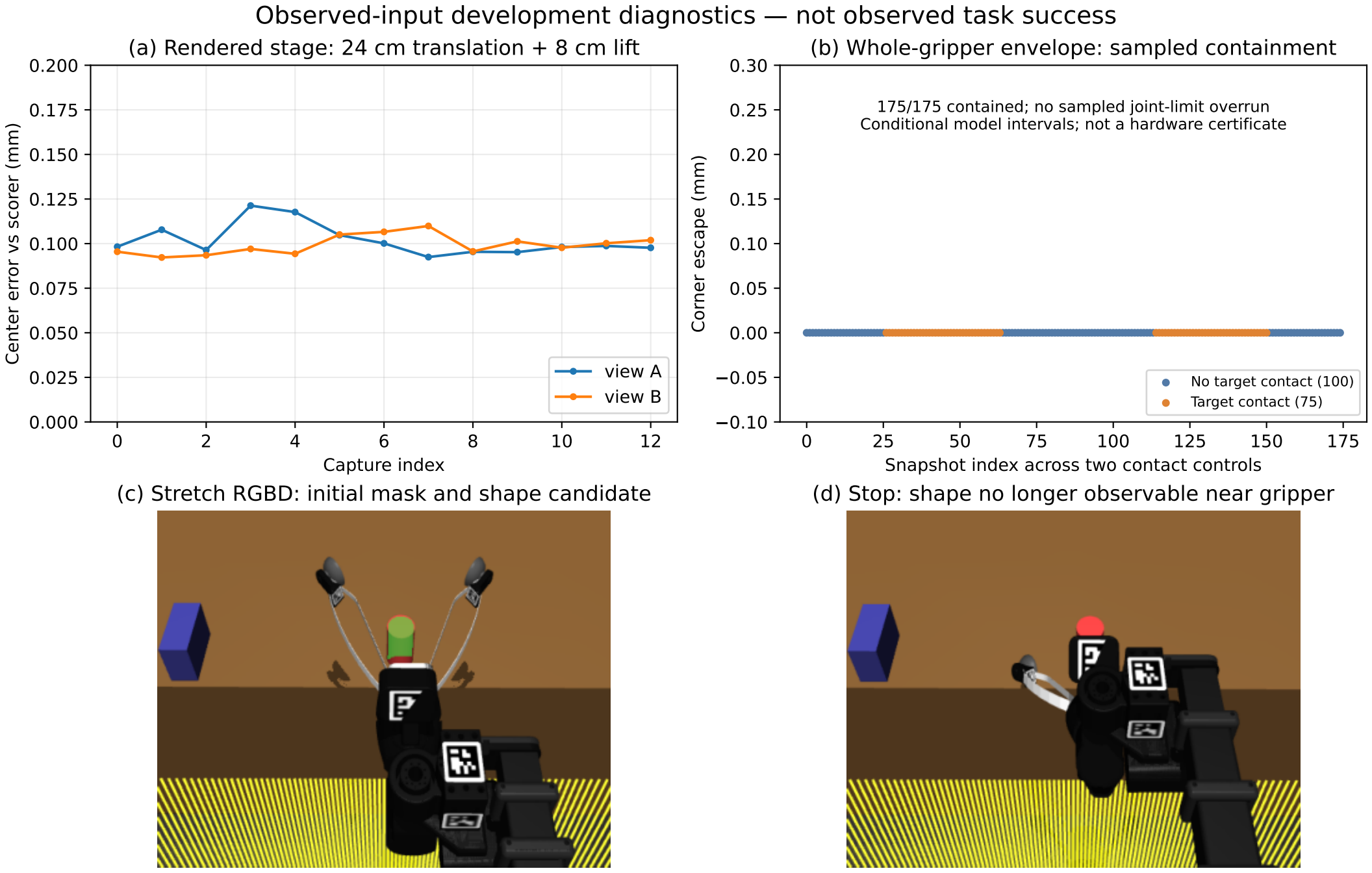

Observed-input development diagnostics — not observed task success
(a) Rendered stage: 24 cm translation + 8 cm lift
Center error vs scorer (mm)
Capture index
view A
view B
(b) Whole-gripper envelope: sampled containment
175/175 contained; no sampled joint-limit overrun
Conditional model intervals; not a hardware certificate
Corner escape (mm)
Snapshot index across two contact controls
No target contact (100)
Target contact (75)
(c) Stretch RGBD: initial mask and shape candidate
(d) Stop: shape no longer observable near gripper